



Department of CSE, PES University

UE24CS341A: SOFTWARE ENGINEERING
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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
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Software Requirements Specification (SRS) of Web Based ML Model Evaluator

Project: Web Based ML Model Evaluator

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1. Introduction

1.1 Purpose

This document is a **Software Requirements Specification (SRS)** for the **Web Based ML Model Evaluator**. It defines the functional and non-functional requirements, system interfaces, features, and verification criteria of the proposed web application.

The system is designed to help users upload datasets, prepare the data, select suitable machine-learning models, train and test the models, evaluate their performance using appropriate metrics, and compare the results through a web-based interface.

1.2 Scope

The Web Based ML Model Evaluator covers the complete basic workflow of machine-learning model evaluation through a web application. It includes **dataset upload and validation, dataset preview, data preprocessing, target-column selection, classification or regression task selection, train-test splitting, model selection, model training, model testing, performance evaluation, result comparison, and report generation**.

The system primarily focuses on **tabular datasets and supervised-learning tasks**, such as classification and regression. It allows users to evaluate multiple supported models and compare their performance using suitable evaluation metrics.

The initial scope **does not include deep-learning model development, image or video processing, real-time data evaluation, or production deployment of trained models**.

1.3 Audience

This SRS is intended for the following users and stakeholders:

- **Project Developers:** To understand the system requirements and implement the application.
- **ML Engineers / Academic Users:** To use the platform for model training and evaluation.
- **QA Engineers:** To verify the functional and non-functional requirements through test cases.
- **Project Guide:** To review the proposed system, workflow, and implementation.
- **Course Coordinator / Assessment Evaluators:** To evaluate whether the project satisfies the required software engineering objectives.

1.4 Definitions

The following terms and abbreviations are used throughout this document:

Term	Definition
ML	Machine Learning, a method of enabling systems to learn patterns from data.
SRS	Software Requirements Specification, a document describing the requirements of the software system.
UI	User Interface through which users interact with the web application.
API	Application Programming Interface used for communication between software components.
CSV	Comma-Separated Values, a common format used for storing tabular datasets.
Dataset	A collection of data used for training and evaluating machine-learning models.
Feature	An input variable or attribute used by a machine-learning model.
Target	The output variable that the model is trained to predict.
Classification	A supervised-learning task in which the model predicts a category or class.
Regression	A supervised-learning task in which the model predicts a numerical value.
Training	The process of learning patterns from the training dataset to build a machine-learning model.
Testing	The process of evaluating a trained model using previously unseen test data.
Preprocessing	Preparing raw data for machine learning by handling missing values, encoding categorical data, scaling features, and performing other required transformations.
Evaluation Metric	A measure used to determine the performance of a trained machine-learning model, such as accuracy, F1-score, MAE, RMSE, or R ² .

2. Overall Description

2.1 Product Perspective

The **Web Based ML Model Evaluator** is a browser-based application that provides a complete workflow for evaluating machine-learning models. Users interact with the system through a web interface to upload datasets, prepare the data, select machine-learning models, train and test them, and view their performance.

The system consists of a **web interface, backend application, data-processing pipeline, model-selection and training modules, evaluation module, result-visualization component, and database storage**. The backend uses machine-learning and data-processing libraries to perform the required operations and communicates with the frontend through application interfaces.

2.2 Major Product Functions (Detailed)

The major functions of the system are:

- User registration, login, and logout.
- Creation and management of evaluation projects.
- Dataset upload and file validation.
- Dataset preview and basic data analysis.
- Selection of the target column and ML task type.
- Data preprocessing, including missing-value handling, encoding, and feature scaling.
- Splitting the dataset into training and testing data.
- Selection of suitable classification and regression models.
- Configuration of selected model parameters.
- Training of selected machine-learning models.
- Testing trained models using unseen test data.
- Calculation of appropriate evaluation metrics.
- Comparison of multiple model performances using tables and charts.
- Storage and viewing of previous evaluation results.
- Generation of downloadable evaluation reports.
- Display of training, testing, and processing status.
- Handling and displaying errors during data processing, training, and evaluation.

2.3 User Roles and Characteristics (Expanded)

Registered User: A user with basic or intermediate knowledge of machine learning. The user can upload datasets, configure preprocessing, select models, run training and evaluation, compare results, and generate reports.

Project Guide / Evaluator: A project guide or evaluator who reviews the system workflow, implementation, requirements, and model-evaluation results for academic assessment.

Administrator: The administrator manages users, evaluation projects, system records, and operational logs. The administrator may also monitor the overall functioning of the application.

2.4 Operating Environment

The system will operate through modern web browsers such as **Google Chrome, Mozilla Firefox, and Microsoft Edge**.

The proposed implementation environment includes:

- **Backend:** Python with Django.
- **Data Processing:** Pandas and NumPy.
- **Machine Learning:** scikit-learn.
- **Frontend:** HTML, CSS, JavaScript, and Bootstrap, or a suitable web frontend.
- **Visualization:** Chart.js or Plotly.js.
- **Database:** SQLite or PostgreSQL.
- **Background Processing:** Celery with Redis, if required for longer training tasks.
- **Client:** A computer or tablet with a modern web browser and a stable network connection.
- **Server:** A multi-core system with sufficient RAM and storage for datasets and model-related files.

2.5 Constraints

The system will initially focus on **tabular datasets, primarily in CSV format**, and supervised-learning tasks such as classification and regression.

The size of uploaded datasets will be limited according to the available server resources. Only machine-learning models and preprocessing techniques supported by the selected libraries will be available through the system.

The system must validate uploaded datasets and user inputs before processing. User accounts, uploaded datasets, model results, and other stored information must be protected from unauthorized access.

The initial version of the system does not cover **deep-learning model development, image/video processing, real-time data evaluation, or production deployment of trained models**.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a **responsive web-based user interface** through which users can perform the complete model-evaluation workflow.

The main interfaces shall include:

- **Login and Registration:** Allows users to create accounts and securely access the system.
- **Dashboard:** Displays evaluation projects and provides options to create or manage projects.
- **Dataset Upload Interface:** Allows users to upload datasets and view file-validation status.
- **Dataset Preview Interface:** Displays dataset rows, column names, data types, and basic dataset information.
- **Data Preprocessing Interface:** Allows users to select the target column, task type, missing-value handling, encoding, scaling, and train-test split.
- **Model Selection Interface:** Allows users to select supported classification or regression models and configure important parameters.
- **Training and Testing Interface:** Displays the status and progress of model training and testing.
- **Results Dashboard:** Displays evaluation metrics, tables, and charts for comparing selected models.
- **Report Interface:** Allows users to generate and download evaluation reports.

The interface shall provide clear labels, understandable messages, keyboard navigation where applicable, and sufficient colour contrast for important information and status indicators.

3.2 Hardware Interfaces

No specialised hardware is required for normal operation of the system.

Client-side hardware

- A desktop computer, laptop, or tablet.
- Minimum **4 GB RAM** recommended.
- A modern web browser.
- Stable network connectivity.

Server-side hardware

- A multi-core CPU system capable of running the web application and machine-learning processes.
- Sufficient RAM for handling uploaded datasets and model training.
- Adequate storage for datasets, model artifacts, project information, and evaluation results.
- GPU support may be used optionally if larger computational workloads require it.

3.3 Software Interfaces

The system shall interact with the following software components:

- **Web Application Backend:** Django with Python for application logic and server-side processing.
- **Data Processing:** Pandas and NumPy for dataset loading, analysis, and preprocessing.
- **Machine Learning:** scikit-learn for supported classification and regression models, training, testing, and evaluation.
- **Frontend:** HTML, CSS, JavaScript, and Bootstrap or another suitable web-interface framework.
- **Database:** SQLite or PostgreSQL for storing user information, projects, dataset metadata, model runs, and evaluation results.
- **Visualization:** Chart.js or Plotly.js for displaying model-performance comparisons through charts.
- **Background Processing:** Celery with Redis may be used to execute longer model-training and evaluation tasks without blocking the web interface.
- **File Storage:** Used for storing uploaded datasets and generated model artifacts where required.

The different software components shall communicate through defined application interfaces and exchange structured data in appropriate formats.

3.4 Communications

- All communication between the user's browser and the web application shall use **HTTPS with TLS 1.2 or higher**.
- The application shall use **HTTP/HTTPS-based requests** for communication between the frontend and backend.
- Data exchanged through application APIs shall use **JSON** where applicable.
- Dataset files shall be transferred using an appropriate multipart file-upload mechanism.
- The system shall provide processing-status updates to the user while longer training or evaluation operations are running.
- Appropriate authentication and session mechanisms shall be used to prevent unauthorized access to user projects, datasets, and evaluation results.
- Communication failures shall be handled with suitable error messages without exposing sensitive system information.

4. System Features (Detailed)

Each requirement below defines a specific function that the **Web Based ML Model Evaluator** shall provide. Every requirement includes its type, priority, source/stakeholder, acceptance criteria, and corresponding test-case reference.

Requirement IDs follow the format **ML-F-###**, where **F** represents a functional requirement.

4.1 Web Interface & Account Management

Description: Provide the web-based entry point to the system, including user registration, authentication, session management, and evaluation-project management.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-001	The system shall allow a user to register using an email address and password.	Functional	High	Registered User	AC-ML-F-001: A valid registration creates a user account. Test: TC-Auth-01	Requires user database
ML-F-002	The system shall authenticate registered users using valid login credentials.	Functional	High	Registered User	AC-ML-F-002: Valid credentials allow access; invalid credentials are rejected. Test: TC-Auth-02	Authentication module required
ML-F-003	The system shall allow an authenticated user to create, rename, and delete evaluation projects.	Functional	High	Registered User	AC-ML-F-003: Project changes are reflected in the user's project list. Test: TC-Proj-01	Project management module
ML-F-004	The system shall allow a user to securely log out of the application and terminate the active session.	Functional	Medium	Registered User	AC-ML-F-004: After logout, protected pages cannot be accessed without authentication. Test: TC-Auth-03	Session management required

4.2 Dataset Upload & Data Pipeline

Description: Accept, validate, inspect, and preprocess tabular datasets and prepare the data for model training and testing.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-010	The system shall allow users to upload datasets in CSV format up to a configurable maximum file size.	Functional	High	Registered User	AC-ML-F-010: Valid files within the configured limit are accepted and stored; oversized files are rejected. Test: TC-Data-01	File storage required
ML-F-011	The system shall validate the uploaded dataset for a valid structure, including a non-empty file, header row, and parsable data.	Functional	High	Registered User	AC-ML-F-011: Invalid or corrupted datasets are rejected with a descriptive error. Test: TC-Data-02	Pandas/file validation
ML-F-012	The system shall display a preview of the uploaded dataset including initial rows, column names, inferred data types, and basic statistics.	Functional	Medium	Registered User	AC-ML-F-012: Dataset preview displays the required information correctly. Test: TC-Data-03	Uses Pandas
ML-F-013	The system shall allow the user to select the target column and choose between classification and regression tasks.	Functional	High	Registered User	AC-ML-F-013: Selected target column and task type are stored for the evaluation. Test: TC-Data-04	Uses scikit-learn

ML-F-014	The system shall allow users to configure preprocessing options including missing-value handling, categorical encoding, feature scaling, and train-test split ratio.	Functional	High	Registered User	AC-ML-F-014: Configured preprocessing is applied and the resulting training and testing datasets are generated correctly. Test: TC-Data-05	Preprocessing pipeline required
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4.3 Model Selection

Description: Provide a catalogue of supported machine-learning models and allow users to select and configure models suitable for their chosen task.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-020	The system shall provide a catalogue of supported classification models, including Logistic Regression, Decision Tree, Random Forest, SVM, k-NN, and Naive Bayes.	Functional	High	ML User	AC-ML-F-020: All supported classification models are displayed when classification is selected. Test: TC-Model-01	scikit-learn
ML-F-021	The system shall provide a catalogue of supported regression models, including Linear Regression, Ridge, Lasso, Decision Tree Regressor, and Random Forest Regressor.	Functional	High	ML User	AC-ML-F-021: All supported regression models are displayed when regression is selected. Test: TC-Model-02	scikit-learn

ML-F-022	The system shall allow users to select one or more compatible models for evaluation and comparison.	Functional	High	ML User	AC-ML-F-022: Multiple compatible models can be selected for a single evaluation project. Test: TC-Model-03	Model-selection interface
ML-F-023	The system shall allow users to configure supported model hyperparameters using valid values and provide suitable default values.	Functional	Medium	ML User	AC-ML-F-023: Valid parameter values are accepted and invalid values are rejected. Test: TC-Model-04	Depends on supported models

4.4 Model Training

Description: Train the selected machine-learning models using the prepared training dataset and provide information about the training process.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-030	The system shall train each selected model using the prepared training dataset.	Functional	High	ML User	AC-ML-F-030: Each selected compatible model completes training successfully or reports an appropriate error. Test: TC-Train-01	scikit-learn
ML-F-031	The system shall execute model-training tasks in a way that does not unnecessarily block the user's web session.	Functional	High	ML User	AC-ML-F-031: The user interface remains accessible while a training task is running. Test: TC-Train-02	Background processing may be used

ML-F-032	The system shall display the current status or progress of an ongoing training task.	Functional	Medium	ML User	AC-ML-F-032: Training status is displayed until the task completes or fails. Test: TC-Train-03	UI/status mechanism required
ML-F-033	The system shall store the trained model information along with its parameters, training timestamp, and execution duration.	Functional	High	ML User / Administrator	AC-ML-F-033: Completed model-run information can be retrieved from the corresponding project. Test: TC-Train-04	Database/model storage required

4.5 Model Testing & Evaluation

Description: Test trained models using unseen test data and calculate appropriate performance metrics according to the selected machine-learning task.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-040	The system shall evaluate each trained model using the held-out test dataset.	Functional	High	ML User	AC-ML-F-040: Each successfully trained model produces an evaluation result using the test dataset. Test: TC-Eval-01	Testing pipeline required

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ML-F-041	For classification tasks, the system shall calculate applicable metrics such as accuracy, precision, recall, F1-score, and confusion matrix.	Functional	High	ML User	AC-ML-F-041: Applicable classification metrics are calculated and displayed correctly. Test: TC-Eval-02	Depends on task type
ML-F-042	For regression tasks, the system shall calculate applicable metrics such as MAE, RMSE, and R^2 .	Functional	High	ML User	AC-ML-F-042: Applicable regression metrics are calculated and displayed correctly. Test: TC-Eval-03	Depends on task type
ML-F-043	The system shall store evaluation results and associate them with the corresponding project and model run.	Functional	High	ML User	AC-ML-F-043: Stored results can be retrieved from the relevant project and model run. Test: TC-Eval-04	Database required

4.6 Results Comparison & Reporting

Description: Present evaluation results in an understandable form and allow users to compare models and generate reports.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
ML-F-050	The system shall display a side-by-side comparison of the evaluation metrics of selected models within a project.	Functional	High	ML User	AC-ML-F-050: Metrics for selected models are displayed together for comparison. Test: TC-Result-01	Results module
ML-F-051	The system shall provide graphical visualizations of model-performance metrics using suitable charts.	Functional	Medium	ML User	AC-ML-F-051: Appropriate charts are generated from the stored evaluation results. Test: TC-Result-02	Chart.js/Plotly.js
ML-F-052	The system shall allow users to mark a preferred or best-performing model for an evaluation project.	Functional	Medium	ML User	AC-ML-F-052: The selected preferred model is stored and displayed in the project results. Test: TC-Result-03	Based on user selection

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ML-F-053	The system shall generate a downloadable evaluation report containing the dataset details, selected models, parameters, and evaluation metrics.	Functional	High	ML User	AC-ML-F-053: A report is generated with the required evaluation information and can be downloaded. Test: TC-Report-01	PDF/CSV generation
ML-F-054	The system shall allow users to view previously completed evaluation results associated with their projects.	Functional	Medium	Registered User	AC-ML-F-054: Previously stored evaluation results can be opened from the corresponding project. Test: TC-Result-04	Database required

5. Non-Functional Requirements (Detailed)

The following non-functional requirements define the quality, performance, reliability, usability, security, and maintainability expectations of the **Web Based ML Model Evaluator**. Each requirement is measurable and is associated with an appropriate acceptance test.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
ML-NF-001	The system shall load the main dashboard and commonly used application pages within 3 seconds under normal operating conditions.	Performance	High	90% of page-load requests complete within 3 seconds. Test: TC-Perf-01
ML-NF-002	The system shall display a dataset preview within 3 seconds for datasets up to 10 MB , excluding file-upload time.	Performance	High	Preview is generated within 3 seconds for supported datasets under 10 MB. Test: TC-Perf-02
ML-NF-003	The system shall remain available for at least 99% of the scheduled operating time , excluding planned maintenance.	Reliability	High	Availability monitoring shows at least 99% uptime during the evaluation period. Test: TC-Avail-01
ML-NF-004	The system shall provide clear and understandable error messages when dataset validation, preprocessing, training, or evaluation fails.	Usability	Medium	Each tested failure condition displays an appropriate error message without exposing internal system details. Test: TC-UX-01
ML-NF-005	The system shall support keyboard navigation, descriptive labels, and sufficient colour contrast for major user-interface components.	Usability / Accessibility	Medium	Accessibility checks pass for the major application screens. Test: TC-UX-02
ML-NF-006	The system shall maintain the consistency of stored project, model-run, and evaluation-result information after completed operations.	Data Integrity	High	Stored information matches the corresponding completed operation during verification. Test: TC-Data-06



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ML-NF-007	The system shall be designed using modular components so that individual modules such as data processing, model training, evaluation, and reporting can be maintained independently.	Maintainability	Medium	Individual modules can be modified or tested without requiring changes to unrelated modules. Test: TC-Maint-01
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5.1 Security

Security is important because the system handles **user accounts, uploaded datasets, model information, evaluation results, and generated reports.**

5.1.1 Security Objectives

The system shall follow the following security objectives:

SO-001 – Confidentiality:

Protect user accounts, uploaded datasets, model artifacts, and evaluation results from unauthorized access.

SO-002 – Integrity:

Ensure that datasets, model configurations, evaluation results, and stored project information are not improperly modified or tampered with.

These two objectives directly cover the main security concerns of our application.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref
PRJ-SR-001	The system shall use HTTPS with TLS 1.2 or higher for client-server communication.	Security	High	Network traffic is encrypted and insecure HTTP access is rejected or redirected. Test: TC-Sec-01
PRJ-SR-002	The system shall securely store user passwords using a suitable one-way password-hashing mechanism and shall not store passwords in plaintext.	Security	High	Database inspection confirms that plaintext passwords are not stored. Test: TC-Sec-02

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PRJ-SR-003	The system shall restrict access to projects, datasets, model runs, and evaluation results to authorized users.	Security	High	An authenticated user cannot access another user's protected resources. Test: TC-Sec-03
PRJ-SR-004	The system shall validate uploaded dataset files and reject unsupported, corrupted, or invalid files before processing.	Security	High	Invalid or unsupported files are rejected without being processed. Test: TC-Sec-04
PRJ-SR-005	The system shall validate user inputs and model parameters before processing them.	Security	Medium	Invalid or malicious input values are rejected without causing unintended system behaviour. Test: TC-Sec-05
PRJ-SR-006	The system shall maintain security-related logs for important events such as login attempts, access violations, and administrative actions.	Security / Audit	Medium	Security events are recorded with timestamps and relevant event information without storing sensitive credentials. Test: TC-Sec-06

6. Quality Attributes & Acceptance Tests

The system shall be evaluated against the defined functional, non-functional, and security requirements.

Acceptance testing shall verify that the major system functions operate correctly and that the required quality attributes are satisfied.

Quality Attributes

The system shall demonstrate the following quality attributes:

- **Performance:** The web interface and dataset-processing operations should respond within the defined time limits.
- **Reliability:** The system should perform evaluation operations consistently without unexpected failures.
- **Usability:** Users should be able to complete the dataset-to-evaluation workflow through clear and understandable interfaces.
- **Security:** User accounts, datasets, model information, and evaluation results shall be protected from unauthorized access.
- **Data Integrity:** Stored datasets, model runs, project information, and evaluation results shall remain consistent and accurate.
- **Maintainability:** The system should use modular components so that individual modules can be tested and maintained independently.
- **Accessibility:** Major user-interface components should support keyboard navigation, descriptive labels, and sufficient colour contrast.

Acceptance Tests

The following test suites shall be used to verify the system:

- **Authentication Tests:** Verify registration, login, logout, and access control.
- **Dataset Tests:** Verify dataset upload, file validation, preview, preprocessing, and train-test splitting.
- **Model Selection Tests:** Verify model catalogue, model compatibility, selection, and parameter configuration.
- **Training Tests:** Verify model training, training status, model information storage, and error handling.
- **Evaluation Tests:** Verify testing of trained models and correct calculation of classification and regression metrics.
- **Results & Reporting Tests:** Verify result comparison, visualization, model selection, historical results, and report generation.
- **Performance Tests:** Verify page-response and dataset-preview performance against the defined NFRs.
- **Security Tests:** Verify HTTPS, password protection, authorization, file validation, input validation, and security logging.



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- **Accessibility Tests:** Verify major interface accessibility requirements.

Acceptance Criteria

The project shall be considered acceptable when:

1. All **high-priority functional requirements** are implemented and successfully verified.
2. All defined **security requirements** pass their corresponding security tests.
3. No critical **non-functional requirement** fails during acceptance testing.
4. Dataset processing, model training, testing, evaluation, comparison, and reporting workflows operate successfully.
5. The **Requirements Traceability Matrix (RTM)** shows that the defined requirements have corresponding design specifications, system modules, and test cases.

7. System Models and Diagrams

The system models provide a visual representation of the major interactions between users and the **Web Based ML Model Evaluator**. UML use-case diagrams are used to show the main functions provided by the system and how users interact with those functions.

Two use-case diagrams are provided to represent the major stages of the system workflow. The first diagram focuses on **user, project, and dataset management**, while the second focuses on **machine-learning model evaluation and result reporting**.

7.1 UML Use-Case Diagram 1 – User, Project and Data Management

Figure 7.1: UML Use-Case Diagram – User, Project and Data Management

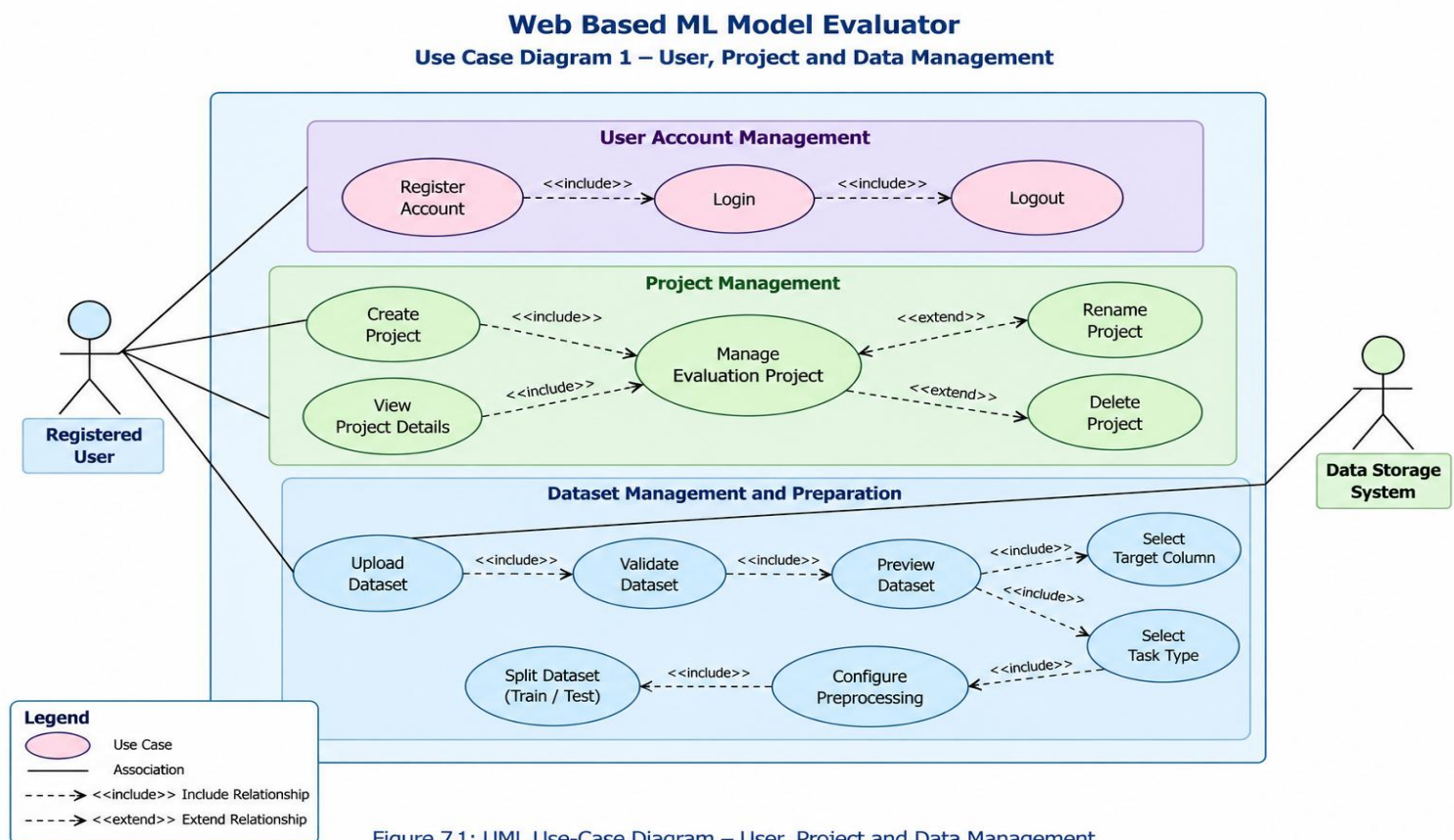


Figure 7.1: UML Use-Case Diagram – User, Project and Data Management

This use-case diagram represents the initial stage of the Web Based ML Model Evaluator. The **Registered User** interacts with the system to register, log in, manage evaluation projects, and upload datasets. The diagram also represents the main data-preparation activities performed by the user, including **dataset validation, dataset**

preview, target-column selection, task-type selection, preprocessing configuration, and train-test splitting. These activities prepare the dataset for the model-selection and evaluation stages.

7.2 UML Use-Case Diagram 2 – Model Evaluation and Results

Figure 7.2: UML Use-Case Diagram – Model Evaluation and Results

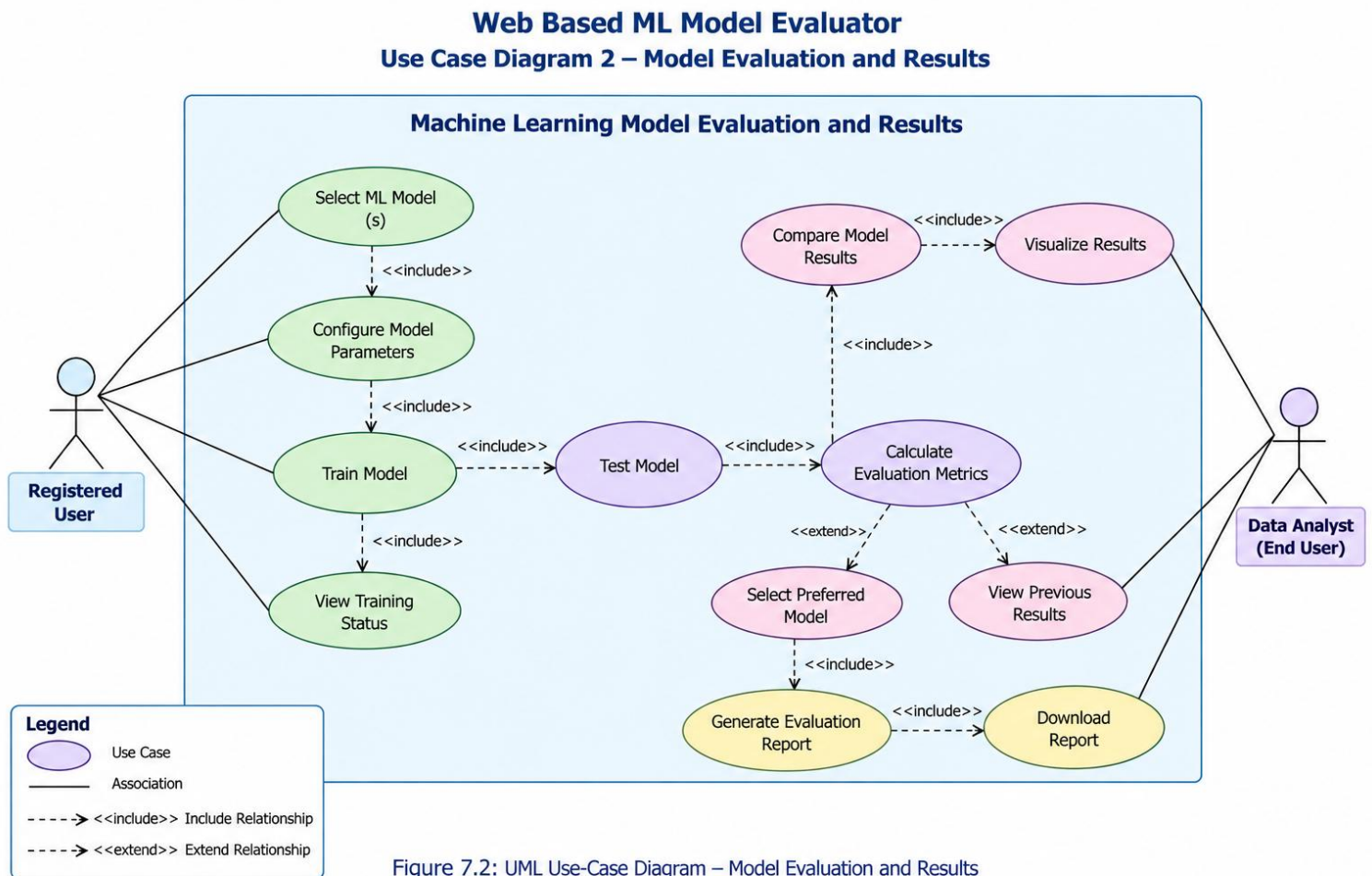


Figure 7.2: UML Use-Case Diagram – Model Evaluation and Results

This use-case diagram represents the machine-learning evaluation stage of the system. The **Registered User** selects one or more suitable machine-learning models, configures their parameters, and starts the model-training process. The system then performs testing and calculates the appropriate evaluation metrics based on the selected task type. The diagram also covers the final result-handling activities, including **comparison of model performances, result visualization, selection of a preferred model, viewing previous results, and generation and download of evaluation reports**. This represents the complete workflow from trained models to the final evaluation output.

8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) maps the functional, non-functional, and security requirements of the **Web Based ML Model Evaluator** to their corresponding design specifications, system modules, and test cases. It ensures that each defined requirement can be traced through the design and implementation stages and verified through testing.

Status codes:

- **N** – Not yet implemented
- **P** – Partially implemented
- **A** – Implemented and accepted

8.1 Functional Requirements Traceability

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
ML-F-001	Register user account	4.1 / DS-Auth-01	Authentication Module	TC-Auth-01	N	User registration required
ML-F-002	Authenticate user	4.1 / DS-Auth-02	Authentication Module	TC-Auth-02	N	Valid credentials required
ML-F-003	Manage evaluation projects	4.1 / DS-Proj-01	Project Management Module	TC-Proj-01	N	User-owned projects
ML-F-004	Logout and terminate session	4.1 / DS-Auth-03	Authentication Module	TC-Auth-03	N	Session invalidation required
ML-F-010	Upload CSV dataset	4.2 / DS-Data-01	Dataset Module	TC-Data-01	N	File storage required
ML-F-011	Validate dataset structure	4.2 / DS-Data-02	Dataset Validation Module	TC-Data-02	N	Invalid files rejected
ML-F-012	Preview dataset	4.2 / DS-Data-03	Data Analysis Module	TC-Data-03	N	Uses Pandas
ML-F-013	Select target and task type	4.2 / DS-Data-04	Data Configuration Module	TC-Data-04	N	Classification / regression

ML-F-014	Configure preprocessing and split	4.2 / DS-Data-05	Preprocessing Module	TC-Data-05	N	Uses scikit-learn
ML-F-020	Provide classification models	4.3 / DS-Model-01	Model Selection Module	TC-Model-01	N	Supported models required
ML-F-021	Provide regression models	4.3 / DS-Model-02	Model Selection Module	TC-Model-02	N	Supported models required
ML-F-022	Select multiple models	4.3 / DS-Model-03	Model Selection Module	TC-Model-03	N	Models must be compatible
ML-F-023	Configure model parameters	4.3 / DS-Model-04	Model Configuration Module	TC-Model-04	N	Validate parameter values
ML-F-030	Train selected models	4.4 / DS-Train-01	Training Module	TC-Train-01	N	Uses prepared training data
ML-F-031	Run training without blocking UI	4.4 / DS-Train-02	Background Training Module	TC-Train-02	N	Background processing
ML-F-032	Display training status	4.4 / DS-Train-03	Training Status Module	TC-Train-03	N	Status update required
ML-F-033	Store model-run information	4.4 / DS-Train-04	Model Storage Module	TC-Train-04	N	Database/storage required
ML-F-040	Test trained models	4.5 / DS-Eval-01	Testing Module	TC-Eval-01	N	Uses held-out test data
ML-F-041	Calculate classification metrics	4.5 / DS-Eval-02	Evaluation Module	TC-Eval-02	N	Accuracy, precision, recall, F1
ML-F-042	Calculate regression metrics	4.5 / DS-Eval-03	Evaluation Module	TC-Eval-03	N	MAE, RMSE, R ²
ML-F-043	Store evaluation results	4.5 / DS-Eval-04	Results Storage Module	TC-Eval-04	N	Linked to project and model run

ML-F-050	Compare model results	4.6 / DS-Result-01	Results Comparison Module	TC-Result-01	N	Side-by-side comparison
ML-F-051	Visualize evaluation results	4.6 / DS-Result-02	Visualization Module	TC-Result-02	N	Charts required
ML-F-052	Select preferred model	4.6 / DS-Result-03	Results Module	TC-Result-03	N	Store user selection
ML-F-053	Generate evaluation report	4.6 / DS-Report-01	Reporting Module	TC-Report-01	N	PDF/CSV report
ML-F-054	View previous evaluation results	4.6 / DS-Result-04	Results History Module	TC-Result-04	N	Database retrieval required

8.2 Non-Functional Requirements Traceability

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
ML-NF-001	Dashboard/page response $\leq$ 3 seconds	5 / DS-Perf-01	Web Interface	TC-Perf-01	N	90% target
ML-NF-002	Dataset preview $\leq$ 3 seconds for datasets up to 10 MB	5 / DS-Perf-02	Data Analysis Module	TC-Perf-02	N	Upload time excluded
ML-NF-003	99% scheduled availability	5 / DS-Avail-01	System / Backend	TC-Avail-01	N	Planned maintenance excluded
ML-NF-004	Clear processing and error messages	5 / DS-UX-01	User Interface	TC-UX-01	N	No sensitive internal details

ML-NF-005	Keyboard navigation and accessible interface	5 / DS-UX-02	User Interface	TC-UX-02	N	Major UI components
ML-NF-006	Maintain project and result data integrity	5 / DS-Data-06	Database / Data Module	TC-Data-06	N	Stored data must remain consistent
ML-NF-007	Modular and maintainable architecture	5 / DS-Maint-01	System Architecture	TC-Maint-01	N	Independent module maintenance

8.3 Security Objectives Traceability

Objective ID	Security Objective	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
SO-001	Protect user accounts, datasets, model artifacts, and results from unauthorized access.	5.1.1 / DS-Sec-01	Authentication & Authorization Module	TC-Sec-03	N	Confidentiality
SO-002	Protect datasets, configurations, and evaluation results from unauthorized modification or tampering.	5.1.1 / DS-Sec-02	Security & Data Integrity Module	TC-Sec-04	N	Integrity

8.4 Security Requirements Traceability

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
PRJ-SR-001	Enforce HTTPS with TLS 1.2+	5.1.2 / DS-Sec-03	Communication / Security Module	TC-Sec-01	N	Secure network communication

PRJ-SR-002	Securely hash user passwords	5.1.2 / DS-Sec-04	Authentication Module	TC-Sec-02	N	Plaintext passwords prohibited
PRJ-SR-003	Restrict access to user resources	5.1.2 / DS-Sec-05	Authorization Module	TC-Sec-03	N	User-level access control
PRJ-SR-004	Validate uploaded dataset files	5.1.2 / DS-Sec-06	Dataset Validation Module	TC-Sec-04	N	Reject invalid files
PRJ-SR-005	Validate user inputs and model parameters	5.1.2 / DS-Sec-07	Input Validation Module	TC-Sec-05	N	Prevent unsafe input
PRJ-SR-006	Maintain security audit logs	5.1.2 / DS-Sec-08	Audit / Logging Module	TC-Sec-06	N	Sensitive credentials excluded

8.5 RTM Coverage Summary

Requirement Category	Required	Covered in RTM
Functional Requirements	Minimum 15	26 – Yes
Non-Functional Requirements	Minimum 5	7 – Yes
Security Objectives	Minimum 2	2 – Yes
Security Requirements	Minimum 5	6 – Yes
Test Case References	Required	Yes
Design Specifications	Required	Yes
System Modules	Required	Yes



Department of CSE, PES University

UE24CS341A: SOFTWARE ENGINEERING
5TH SEMESTER, ACADEMIC YEAR 2026-27
SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
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Project Contribution Details

S. No.	Name of the Student	SRN	Contribution
1	Shreyas Bellad	PES2UG25CS824	Introduction and Overall Description
2	Nirupama Jayaraman	PES2UG24CS324	External Interface Requirements and System Features
3	Pratham Kumar	PES2UG24CS368	Non-Functional Requirements and Quality Attributes & Acceptance Tests
4	Parinitha Acharya	PES2UG24CS339	UML Use-Case Diagrams and Requirements Traceability Matrix (RTM)

THANK_YOU